Yugma TechFest 2.0 - MedhaDrishti National-Level AI Hackathon

Hackathon Challenge: AI for Object Recognition

Topic: AI for Multimodal RGB-Thermal Pedestrian Detection through Efficient Fusion Strategies

Theme: Designing Efficient RGB-Thermal Fusion Strategies for Robust Pedestrian Detection

Background:

Pedestrian detection is an important computer vision task with applications in intelligent surveillance, autonomous systems, transportation, disaster response, search and rescue, and smart-city applications. While RGB cameras provide rich appearance and texture information, their performance degrades significantly under poor illumination, adverse weather, and night-time conditions. Thermal cameras complement RGB imagery by capturing heat signatures, enabling more reliable detection under challenging environments.

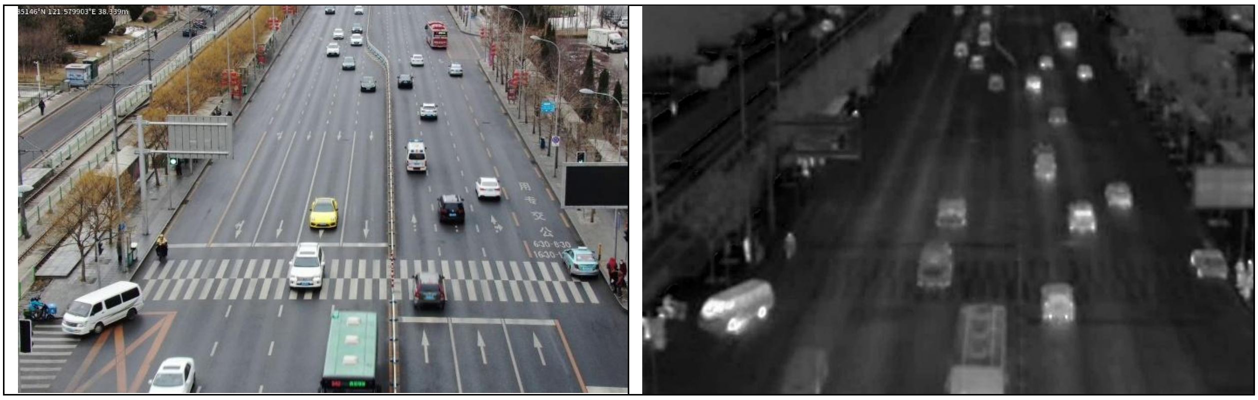

Fig1: Sample paired RGB-Thermal images

Recent advances in RGB-Thermal (RGBT) pedestrian detection have shown that multimodal **fusion significantly improves detection performance. However, designing an effective fusion** strategy that efficiently exploits complementary information from RGB and thermal modalities remains an open research challenge. Furthermore, detecting small and tiny pedestrians is particularly difficult due to limited pixel information, modality imbalance, spatial misalignment, and complex backgrounds. The VTUAV-det benchmark dataset and the Quality-aware RGBT Fusion Detector (QFDet) provide a strong foundation for addressing these challenges.

This hackathon aims to develop efficient RGB-Thermal fusion strategies that enhance pedestrian detection performance, with particular emphasis on improving the detection of small and tiny pedestrians while maintaining computational efficiency.

Problem Statement:

Develop an AI-based RGB-Thermal pedestrian detection framework capable of accurately detecting pedestrians from paired RGB and thermal images by improving the baseline Quality-aware RGBT Fusion Detector (QFDet).

Participants will be provided with the baseline implementation, pretrained weights, and a curated subset of the VTUAV-det dataset. The objective is to design novel RGB-Thermal fusion strategies that improve pedestrian detection performance while maintaining computational efficiency.

Participants are encouraged to improve the baseline by exploring one or more of the following approaches. However, participants are not limited to these approaches and may propose any innovative solution.

- Input-level/Feature-level/Decision-level RGB-Thermal fusion
- Cross-modal attention mechanisms
- Multi-scale feature fusion for small pedestrian detection
- Novel fusion strategies or architectural enhancements

Dataset

A curated subset of the VTUAV-det dataset will be provided.

VTUAV-det is derived from the VTUAV RGB-Thermal tracking benchmark and contains paired RGB and thermal image sequences with pedestrian annotations. The dataset includes diverse scenes captured under varying illumination conditions, viewpoints, weather conditions, and object scales. One of its primary challenges is the detection of small and tiny pedestrians, making it suitable for evaluating robust RGB-Thermal fusion strategies.

The organizers will provide:

- Paired RGB and thermal image pairs
- Pedestrian bounding-box annotations
- Dataset documentation

Dataset Split

Dataset	**Image Pairs**
Training	1200
Validation	300
Test	200

Note: Only the **Pedestrian** class shall be considered.

Hackathon Workflow

The hackathon consists of four sequential stages that guide participants through dataset understanding, preprocessing, baseline evaluation, multimodal fusion development, and performance analysis.

Stage 1: Dataset Exploration, Analysis, and Preparation

Problem Description

Participants shall analyze and prepare the provided VTUAV-det dataset before developing the RGB-Thermal pedestrian detection framework.

The analysis should include:

- Pedestrian instance distribution (total number of pedestrian instances in the training, validation, and test sets, and the average number of pedestrians per image)
- Pedestrian scale distribution:
    - Small: Area $< 32^2$ pixels
    - Medium: $32^2 \leq$ Area $< 96^2$ pixels
    - Large: Area $\geq 96^2$ pixels
- Image resolution
- Characteristics of RGB and thermal images
- Annotation format
- Challenging scenarios (e.g., occlusion, low illumination, clutter, and small pedestrians)
- RGB-Thermal alignment verification by visualizing at least 20 paired RGB-Thermal images with their corresponding annotations displayed side by side.
- Thermal image enhancement (optional).
- Contrast enhancement (optional).

Participants shall also verify the correctness of the provided annotations and RGB-Thermal image alignment. If required, suitable preprocessing techniques such as thermal enhancement or contrast enhancement may be applied with proper justification.

Expected Deliverables

- Pedestrian instance distribution and dataset statistics.
- Pedestrian scale distribution.
- Sample RGB-Thermal image pairs.
- Visualization of annotated RGB-Thermal image pairs.
- RGB-Thermal alignment verification.
- Description and justification of any preprocessing techniques applied.
- Discussion of the key challenges in the VTUAV-det dataset.

Fig 2: Sample annotated paired RGB-Thermal images

Stage 2: Unimodal Analysis and Baseline Performance Benchmarking

Problem Description

Before developing an RGB-Thermal fusion strategy, participants shall analyze the individual performance of RGB and Thermal modalities using the provided baseline detection framework.

The objective is to understand the strengths and limitations of each modality under different imaging conditions and establish a reference for multimodal fusion.

Participants shall evaluate:

- RGB-only detector
- Thermal-only detector

using the provided pretrained baseline weights.

Training from scratch is not permitted. Participants shall use the provided pretrained QFDet weights for evaluation and fine-tuning. Participants shall evaluate the pretrained models on the validation and test datasets separately.

Following the unimodal analysis, participants shall reproduce the provided QFDet baseline and benchmark its RGB-Thermal detection performance.

Expected Deliverables

Participants shall evaluate and compare the RGB-only, Thermal-only, and Baseline QFDet detectors using

- Detection Metrics (COCO object detection evaluation metrics): mAP, mAP50, mAP75, mAPS, mAPM, and mAPL
- Computational Metrics: FPS, Inference Time, Model Size, Number of Parameters, and FLOPs

Participants shall also provide a comparative analysis highlighting the strengths and limitations of each modality**.**

Stage 3: RGB-Thermal Fusion Strategy Development

Problem Description

This is the core stage of the hackathon.

Participants shall improve the baseline QFDet by proposing one or more novel RGB-Thermal fusion strategies.

Possible improvements include, but are not limited to:

Feature Fusion

- Input-level/Feature-level/Decision-level RGB-Thermal fusion
- Cross-modal attention
- Cross-layer feature fusion
- Adaptive feature weighting
- Multi-scale feature aggregation
- Dynamic feature selection

Network Improvements

- Neck redesign
- Detection head modification
- Transformer-based fusion
- Lightweight fusion modules
- Small-object enhancement modules

Learning Improvements

- Loss function

Particular emphasis should be given to improving the detection of small and tiny pedestrians.

Expected Deliverables

- Fusion architecture diagram
- Description of the proposed fusion strategy
- Modified network architecture
- Training methodology used
- Experimental results
- Ablation study (optional)

Stage 4: Performance Evaluation and Comparative Analysis

Problem Description

Participants shall compare the improved detector against the baseline QFDet.

Evaluation shall include both quantitative and qualitative analysis.

Evaluation Metrics

The proposed framework will be evaluated using:

- Detection Performance: COCO metrics (mAP, mAP50, mAP75, mAPS, mAPM, and mAPL)
- Computational Performance: FPS, inference time, model size, number of parameters, and FLOPs
- Qualitative Analysis: Qualitative visualization of detection results, Comparison with the ground truth annotations, and failure case analysis

Expected Deliverables

- Quantitative performance comparison
- Qualitative detection results

Expected Final Deliverables

Each participating team shall submit:

- Source code (ZIP archive or GitHub repository)
- Trained model weights
- Prediction results of validation and test sets (COCO JSON format)
- Technical report (3–5 pages)
- Presentation slides

Evaluation Criteria

Criteria	Weightage
Dataset Exploration, Analysis & Preparation	15%
Unimodal Analysis & Baseline Benchmarking	15%
Novel RGB-Thermal Fusion Strategy Development	40%
Performance Evaluation & Comparative Analysis	20%
Technical Report & Presentation	10%
Total	100%

Hackathon Rules

- Only the Pedestrian class shall be considered.
- Only the provided VTUAV-det subset shall be used.
- Participants must use the supplied pretrained QFDet weights.
- Training from scratch is not permitted.
- External datasets are not allowed.
- Teams may modify any part of the baseline architecture.
- Evaluation has to be done on test set

Resources Provided by Organizers

1. Curated VTUAV-det dataset subset : VTUAV_subset

   https://drive.google.com/drive/folders/1ox8NrnlhreI5-Dnr9yJouT8CbVqaXarJ

2. Baseline QFDet implementation: https://github.com/NNNNerd/mmdet-rgbtdroneperson
3. Pretrained QFDet model weights:
4. Sample training and evaluation scripts:
5. Evaluation scripts.
6. Dataset documentation: https://zhang-pengyu.github.io/DUT-VTUAV/
7. Reference papers and implementations.

Hardware and Software Requirements

Hardware Requirements (any of these)

- Minimum: Intel Core i5 (or equivalent), 12 GB RAM, NVIDIA RTX 3050 (or equivalent GPU with at least 8 GB GPU memory).
- Recommended: Intel Core i7/Ryzen 7 (or higher), 32 GB RAM, NVIDIA RTX 4060 or above (or equivalent GPU with 12–16 GB GPU memory).

- Cloud Computing Platforms : Participants may also use any of the following GPU-enabled cloud platforms:
  - Google Colab (Free, Pro, or Pro+)
  - Kaggle Notebooks (GPU)

  - Institutional GPU servers
  - Other cloud platforms supporting CUDA-enabled GPUs

Software Requirements

- Python 3.10 or later
- PyTorch
- OpenCV
- NumPy
- Torchvision
- MMDetection/OpenMMLab (for the provided baseline)
- Other dependencies listed in the provided requirements.txt
- Colab
- MatLAb

Reference Papers

1. Zhang, Y., Xu, C., Yang, W., He, G., Yu, H., Yu, L., & Xia, G. S. (2023). Drone-based RGBT tiny person detection. *ISPRS Journal of Photogrammetry and Remote Sensing*, *204*, 61-76.

Reference Implementations

1. QFDet (VTUAV-det)
   https://github.com/NNNNerd/mmdet-rgbtdroneperson
2. https://github.com/XinyiYing/RGBT-Tiny
3. MMDetection: https://mmdetection.readthedocs.io/en/latest/get_started.html